BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, AJK, MIRPUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: 21/01

Retyped faithfully from the original board paper — same questions, no changes.

BIOLOGY (PART – I) (Objective Part) — Smart Syllabus

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 21/01

Note: Write your Roll No. in space provided. Over writing, cutting, using of lead pencil will result in loss of marks. All questions are to be attempted.

Q1. National environmental problem in Pakistan is;

(A) Corruption
(B) Pollution
(C) Population
(D) Tanneries

Q2. Lipids are not soluble in;

(A) Chloroform
(B) Ether
(C) Alcohol
(D) Sap

Q3. Some enzymes which are involved in the synthesis of proteins are integral part of;

(A) Lysosome
(B) Nucleolus
(C) Vacuole
(D) Ribosomes

Q4. The size of eukaryotic ribosome is;

(A) 40 S
(B) 70 S
(C) 80 S
(D) 60 S

Q5. Which one is the smallest known virus;

(A) Polio virus
(B) Influenza virus
(C) Herpes virus
(D) Pox virus

Q6. Reserve food material in cyanobacteria is;

(A) Fat
(B) Protein
(C) Sterols
(D) Glycogen

Q7. Giants of the protist kingdom are included in;

(A) Red algae
(B) Brown algae
(C) Diatoms
(D) Green algae

Q8. Histoplasmosis is a serious infection of;

(A) Lungs
(B) Skin
(C) Kidney
(D) Eyes & brain

Q9. Which of the following is a modified leaf;

(A) Root
(B) Fruit
(C) Flower
(D) Shoot

Q10. The excretory system of flatworms is composed of;

(A) Nephridia
(B) Nephron
(C) Kidney
(D) Flame cells

Q11. Earthworm belongs to phylum;

(A) Echinodermata
(B) Annelida
(C) Mollusca
(D) Porifera

Q12. The dark reaction occurs in;

(A) Cytoplasm
(B) Nucleus
(C) Grana
(D) Stroma

Q13. The most common fuel used by the cell to provide energy by cellular respiration is;

(A) Maltose
(B) Glucose
(C) Fructose
(D) Lipid

Q14. Carbohydrate digesting enzyme of pancreas is;

(A) Amylopsin
(B) Lipase
(C) Trypsin
(D) Pepsin

Q15. The air spaces in leaf may comprise up to _____ of the total volume of a leaf;

(A) 80 %
(B) 60 %
(C) 40 %
(D) 20 %

Q16. The normal pH of human blood is;

(A) 7.4
(B) 4.7
(C) 4.5
(D) 5.4

Q17. The cells which supply ATP and proteins to sieve tubes are;

(A) Epidermal
(B) Tracheids
(C) Vessels
(D) Companion

Note: Each question has four possible answers, Tick (√) the correct answer. (17)

BIOLOGY PAPER : PART – I (Subjective Part) — Smart Syllabus

Time Allowed: 2:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 21/01

Note: SECTION – I

Q2. (Marks 16) Write short answers of any eight parts. (2 x 8 = 16)

(i) Write general formula of amino acid.

(ii) What is cofactor? Also write its functions.

(iii) What is active site of an enzyme? Name its parts.

(iv) Differentiate between reversible and irreversible inhibitors.

(v) What is dikaryotic hyphae? How it is formed.

(vi) How endomycorhizae is different from ectomycorhizae?

(vii) What are placental mammals?

(viii) Why sub-phylum vertebrate is also called craniata?

(ix) Write two reasons why Echinodermata is placed at top of list of invertebrates.

(x) Mention two effects of harmful insects.

(xi) What are light dependant reactions and their products?

(xii) What is cyclic phosphorylation? How it is activated?

Q3. (Marks 16) Write short answers of any eight parts. (2 x 8 = 16)

(i) Define parasitology and biotechnology.

(ii) What is hydroponic culture technique?

(iii) Write down the functions of golgi complex.

(iv) Define storage diseases.

(v) What are trichonymphas? Give their role.

(vi) Write down about evolutionary significance of Euglenoids.

(vii) What is sleeping sickness?

(viii) Name four phylum of photosynthetic protists.

(ix) Give any four characteristics of vascular plants, which enable them to become predominant flora of land.

(x) Why bryophytes are said to be amphibian of the plants?

(xi) Define facilitated diffusion.

(xii) What is thalassaemia?

Q4. (Marks 12) Write short answers of any six parts. (2 x 6 = 12)

(i) Write down botanical names of brinjal and potato.

(ii) What are plasmids? Give their function.

(iii) Differentiate between diarrhoea and constipation.

(iv) Define chlorosis. Give its cause.

(v) Write two functions of secretion.

(vi) Give two properties of respiratory surface.

(vii) What are alveoli? Give their function.

(viii) Enlist functions of nasal cavity in man.

(ix) What is myoglobin? Give its function.

> Note: SECTION – II — Note:- Attempt any three questions. (8 x 3 = 24)

Q5. (Marks 8) (a) How scientific problem of biology can be solved by using biological method? Explain.

(b) [04] Write in detail how influx of K^+ ions help in opening and closing of stomata?

Q6. (Marks 8) (a) Discuss primary and secondary structure of proteins.

(b) [04] Describe economic gains due to fungi.

Q7. (Marks 8) (a) Describe the process of nutrition in Bacteria.

(b) [04] Discuss evolution of megaphyll leaf.

Q8. (Marks 8) (a) What is hepatitis? Give its types.

(b) [04] Write a note on citric acid cycle.

Q9. (Marks 8) (a) Write a note on Lysosomes.

(b) [04] Describe symbiotic nutrition in plants and nutrition in insectivorous plants.